

Solution Tutorial IV – Mass Estimation

Task 1

Task 1-4 aim to determine the operating empty mass of the Airbus A320. For every task, compare the results to similar aircraft from the table on next page by building the respective fractions of gross weight.

HINT: Examine the table below. Note that units are in lbm. The expression GW stands for Gross Weight which can be considered to be equivalent to the maximum take-off mass $m_{TO,max}$.

Type	Mc Donnell Douglas DC-9-30	MD-80	DC-10-10	Boeing 737-200	727-100	747-100	Airbus A300-B2
Flight Design Gross Weight, GW, lbs	108,000	140,000	430,000	115,500	160,000	710,000	302,000
Structure/GW	0.286	0.304	0.316	0.270	0.317	0.298	0.353
Power Plant/GW	0.076	0.079	0.077	0.071	0.078	0.062	0.076
Fixed Equipm't/GW	0.175	0.182	0.169	0.129	0.133	0.089	0.116
Σ Empty Weight/GW	0.538	0.564	0.561	0.521	0.552	0.498	0.559
Wing Group/GW	0.106	0.111	0.114	0.092	0.111	0.122	0.146
Empenn. Group/GW	0.026	0.024	0.032	0.024	0.026	0.017	0.020
Fuselage Group/GW	0.103	0.115	0.104	0.105	0.111	0.101	0.119
Nacelle Group/GW	0.013	0.015	0.020	0.012	0.024	0.014	0.023
Land. Gear Group/GW	0.039	0.038	0.046	0.038	0.045	0.044	0.045
Take-off Gross Wht, W_{TO} , lbs	108,000	140,000	430,000	115,500	160,000	710,000	302,000
Empty Weight, W_E , lbs	58,070	78,950	241,250	60,210	88,300	353,398	168,805










Roskam, J. (1985). Airplane design Vol. V, p. 153.

1) Calculate the airframe (structure) mass $m_{airframe}$ of the Airbus A320 in lbm. It consists of:

- Wings (use Torenbeek method)
- Empennage
- Fuselage
- Nacelle & Pylons
- Landing gear

One general remark on the units of the English Engineering Unit System: in many books and papers no clear distinction between lbf (pound-force) and lbm (pound-mass) is made. Instead, often both are named lb or lbs and only from the context it becomes clear if a mass or a force is meant. To make a clear distinction possible, in this tutorial lbf and lbm will be used as appropriate.

- Wing:

Equation to be used (Torenbeek method):

$$m_W = 0.0017 \cdot m_{ZF} \cdot C_W \cdot b_S^{0.75} \cdot \left[1 + \sqrt{\frac{6.25}{b_S}} \right] \cdot n_{z,ult}^{0.55} \cdot \left(\frac{b_S}{t_{max} \cdot \frac{m_{ZF}}{S_{ref}}} \right)^{0.3}$$

Coefficients of equation:

- $m_{ZF} = 61000 \text{ kg} = 134500 \text{ lbm}$
- $C_W = 0.95$ (engine mounted on wings, no strut-braced wings, landing gear on wing)
- $\Lambda_{50\%,W} = \text{atan} \left[\tan(\Lambda_{25\%,W}) - \frac{4}{AR_W} \cdot \left[\frac{50-25}{100} \cdot \frac{1-\lambda_W}{1+\lambda_W} \right] \right] = 0.3815 \text{ rad}$
- $b_S = \frac{b}{\cos \Lambda_{50\%,W}} = \frac{34.1 \text{ m} \cdot 3.2808}{\cos(0.3815)} = 120.5 \text{ ft}$
- $t_{max} = \left(\frac{t}{c} \right)_{root} \cdot c_{root} = 15.15\% \cdot 6.07 \text{ m} \cdot 3.2808 = 3.017 \text{ ft}$
- $n_{z,ult} = +3.75 \text{ g}$ (certification specifications for large aircraft CS-25)
- $S_{ref} = 122.4 \text{ m}^2 = 1317 \text{ ft}^2$

Therefore:

$$m_W = 0.0017 \cdot 134500 \cdot 0.95 \cdot 120.5^{0.75} \cdot \left[1 + \sqrt{\frac{6.25}{120.5}} \right] \cdot 3.75^{0.55} \cdot \left(\frac{120.5}{3.017 \cdot \frac{134500}{1317}} \right)^{0.3}$$

$$= 15140 \text{ lbm}$$

$$\Rightarrow \frac{m_W}{GW} = \frac{15140 \text{ lbm}}{73500 \text{ kg}} = \frac{15140 \text{ lbm}}{162000 \text{ lbm}} = 0.09346$$

- Horizontal tail:

Equation to be used:

$$m_{HT} = C_{HT} \cdot S_{HT} \cdot \left(3.81 \cdot \frac{S_{HT}^{0.2} \cdot V_D}{1000 \cdot \sqrt{\cos \Lambda_{50\%,HT}}} - 0.287 \right)$$

Coefficients of equation:

- $C_{HT} = 1.1$ (variable incidence stabilizer)
- $S_{HT} = 31 \text{ m}^2 = 333.7 \text{ ft}^2$
- $V_D = 381 \text{ kn}$
- $\Lambda_{50\%,HT} = \text{atan} \left[\tan(\Lambda_{25\%,HT}) - \frac{4}{AR_{HT}} \cdot \left[\frac{50-25}{100} \cdot \frac{1-\lambda_{HT}}{1+\lambda_{HT}} \right] \right] = 0.4110 \text{ rad}$

Therefore:

$$m_{HT} = 1.1 \cdot 333.7 \cdot \left(3.81 \cdot \frac{333.7^{0.2} \cdot 381}{1000 \cdot \sqrt{\cos(0.4110)}} - 0.287 \right) = 1673 \text{ lbm}$$

- Vertical tail:

Equation to be used:

$$m_{VT} = C_{VT} \cdot S_{VT} \cdot \left(3.81 \cdot \frac{S_{VT}^{0.2} \cdot V_D}{1000 \cdot \sqrt{\cos \Lambda_{50\%,VT}}} - 0.287 \right)$$

Coefficients of equation:

- $C_{VT} = 1$ (horizontal tail not mounted on vertical tail)
- $S_{VT} = 21.5 \text{ m}^2 = 231.4 \text{ ft}^2$
- $\Lambda_{50\%,VT} = \text{atan} \left[\tan(\Lambda_{25\%,VT}) - \frac{4}{AR_{VT}} \cdot \left[\frac{50-25}{100} \cdot \frac{1-\lambda_{VT}}{1+\lambda_{VT}} \right] \right] = 0.3637 \text{ rad}$

Therefore:

$$m_{VT} = 1 \cdot 231.4 \cdot \left(3.81 \cdot \frac{231.4^{0.2} \cdot 381}{1000 \cdot \sqrt{\cos(0.3637)}} - 0.287 \right) = 965.9 \text{ lbm}$$

- Empennage:

$$m_{EMP} = m_{HT} + m_{VT} = 2639 \text{ lbm}$$

$$\Rightarrow \frac{m_{EMP}}{GW} = \frac{2639 \text{ lbm}}{162000 \text{ lbm}} = 0.01629$$

- Fuselage:

Equation to be used:

$$m_{fus} = 0.021 \cdot C_{fus} \cdot \sqrt{V_D \cdot \frac{d_{HT}}{w_{fus} + H_{fus}}} \cdot S_{fus}^{1.2}$$

Coefficients of equation:

- $C_{fus} = 1.08$ (pressurized cabin, main landing gear & engines not attached to fuselage)
- $w_{fus} = 3.95 \text{ m} = 13.00 \text{ ft}$
- $H_{fus} = 4.14 \text{ m} = 13.58 \text{ ft}$
- $l_{fus} = 37.57 \text{ m} = 123.3 \text{ ft}$
- $S_{fus} \approx l_{fus} \cdot \pi \cdot \left(\frac{w_{fus} + H_{fus}}{2} \right) = 123 \cdot \pi \cdot \left(\frac{13.00 + 13.57}{2} \right) = 5139 \text{ ft}^2$
(crude approximation)
- $d_{HT} \approx x_{0,HT} - x_{0,W} = 31.55 \text{ m} - 11.76 \text{ m} = 64.93 \text{ ft}$
- $V_D = 381 \text{ kn}$

Therefore:

$$m_{fus} = 0.021 \cdot 1.08 \cdot \sqrt{381 \cdot \frac{64.93}{13.00 + 13.58}} \cdot 5139^{1.2} = 19650 \text{ lbm}$$

$$\Rightarrow \frac{m_{fus}}{GW} = \frac{19650 \text{ lbm}}{162000 \text{ lbm}} = 0.1213$$

- Nacelles & Pylons:

Equation to be used:

$$m_{nac} = 0.065 \cdot T_{TO}$$

Coefficients of equation:

- $T_{TO} = T_{eng} \cdot n_{eng} = 2 \cdot 118 \text{ kN} = 236 \text{ kN} = 53050 \text{ lbf}$ (total take-off thrust)

Therefore:

$$m_{nac} = 0.065 \cdot 53050 = 3448 \text{ lbm}$$

$$\Rightarrow \frac{m_{nac}}{GW} = \frac{3448 \text{ lbm}}{162000 \text{ lbm}} = 0.02128$$

- Nose landing gear:

Equation to be used:

$$m_{LG} = C_{LG} \cdot (C_1 + C_2 \cdot m_{TO,max}^{0.75} + C_3 \cdot m_{TO,max} + C_4 \cdot m_{TO,max}^{1.5})$$

Coefficients of equation:

- $C_1 = 20 \mid C_2 = 0.10 \mid C_3 = 0 \mid C_4 = 2 \cdot 10^{-6}$ (nose landing gear factors)
- $C_{LG} = 1$ (low wing configuration)
- $m_{TO,max} = 73500 \text{ kg} = 162000 \text{ lbm}$

Therefore:

$$\begin{aligned} m_{LG,Nose} &= 1 \cdot (20 + 0.10 \cdot 162000^{0.75} + 0 \cdot 162000 + 2 \cdot 10^{-6} \cdot 162000^{1.5}) \\ &= 958.1 \text{ lbm} \end{aligned}$$

- Main landing gear:

Equation to be used:

$$m_{LG} = C_{LG} \cdot (C_1 + C_2 \cdot m_{TO,max}^{0.75} + C_3 \cdot m_{TO,max} + C_4 \cdot m_{TO,max}^{1.5})$$

Coefficients of equation:

- $C_1 = 40 \mid C_2 = 0.16 \mid C_3 = 0.019 \mid C_4 = 1.5 \cdot 10^{-5}$ (main landing gear factors)
- $C_{LG} = 1$ (low wing configuration)

Therefore:

$$\begin{aligned} m_{LG,Main} &= 1 \cdot (40 + 0.16 \cdot 162000^{0.75} + 0.019 \cdot 162000 + 1.5 \cdot 10^{-5} \cdot 162000^{1.5}) \\ &= 5389 \text{ lbm} \end{aligned}$$

- Landing gear:

$$m_{LG} = m_{LG,Nose} + m_{LG,Main} = 6347 \text{ lbm}$$

$$\Rightarrow \frac{m_{LG}}{GW} = \frac{6347 \text{ lbm}}{162000 \text{ lbm}} = 0.03918$$

- Overall:

$$\begin{aligned}
 m_{airframe} &= m_W + m_{EMP} + m_{fus} + m_{nac} + m_{LG} \\
 \Rightarrow m_{airframe} &= 15140 \text{ lbm} + 2639 \text{ lbm} + 19560 \text{ lbm} + 3448 \text{ lbm} + 6347 \text{ lbm} \\
 &= 47220 \text{ lbm}
 \end{aligned}$$

$$\Rightarrow \frac{m_{airframe}}{GW} = \frac{47220 \text{ lbm}}{162000 \text{ lbm}} = 0.2915$$

Task 2

Then also calculate the propulsion mass m_{prop} .

- Propulsion/Engines:

Equation to be used:

$$m_{prop} = 1.15 \cdot C_{TR} \cdot n_{eng} \cdot m_{eng}$$

Coefficients of equation:

- $C_{TR} = 1.18$ (thrust reversal equipped)
- $n_{eng} = 2$ (from three view of aircraft)
- $m_{eng} = 2366 \text{ kg} = 5216 \text{ lbm}$

Therefore:

$$m_{prop} = 1.15 \cdot 1.18 \cdot 2 \cdot 5216 = 14160 \text{ lbm}$$

$$\Rightarrow \frac{m_{prop}}{GW} = \frac{14100 \text{ lbm}}{162000 \text{ lbm}} = 0.08741$$

Task 3

Determine the systems mass m_{sys} (by assuming the aircraft serves short-range routes).

- Systems:

Equation to be used:

$$m_{sys} \approx C_{sys} \cdot m_{TO,max}$$

Coefficients of equation:

- $C_{sys} = 14\%$ (short range duties)
- $m_{TO,max} = 73500 \text{ kg} = 162000 \text{ lbm}$

Therefore:

$$m_{sys} = 14\% \cdot 162000 = 22680 \text{ lbm}$$

Task 4

Finally calculate the operating empty mass m_{OE} .

- Operating empty mass:

$$m_{OE} = m_{airframe} + m_{prop} + m_{sys} = 47260 \text{ lbm} + 14160 \text{ lbm} + 22680 \text{ lbm} \\ = 84100 \text{ lbm}$$

$$\Rightarrow \frac{m_{OE}}{GW} = \frac{84100 \text{ lbm}}{162000 \text{ lbm}} = 0.5191$$

Task 4

The certification specifications for civil aircraft postulate, that the aircraft structure must withstand vertical load factors up to $+3.75 \text{ g}$. A novel flight control system now enables a limitation to $+3.0 \text{ g}$ thanks to dynamic control surface deflections. Due to the reduced peak stress, the wings can therefore be built lighter (assuming the certification specifications change). How much do they weigh in % relation to the original wings?

- Wing:

Equation to be used (Torenbeek method) :

$$m_W = 0.0017 \cdot m_{ZF} \cdot C_W \cdot b_s^{0.75} \cdot \left[1 + \sqrt{\frac{6.25}{b_s}} \right] \cdot n_{z,ult}^{0.55} \cdot \left(\frac{b_s}{t_{max} \cdot \frac{m_{ZF}}{S_{ref}}} \right)^{0.3}$$

Changed coefficients of equation (others remain as in task 1):

- $n_{z,ult} = +3.0 \text{ g}$

Therefore:

$$m_{W,light} = 0.0017 \cdot 134500 \cdot 0.95 \cdot 120.5^{0.75} \cdot \left[1 + \sqrt{\frac{6.25}{120.5}} \right] \cdot 3^{0.55} \\ \cdot \left(\frac{120.5}{3.017 \cdot \frac{134500}{1317}} \right)^{0.3} = 13390 \text{ lbm}$$

- Comparison in relation to original wings:

$$\frac{m_{W,light}}{m_{W,task1}} = \frac{13390 \text{ lbm}}{15140 \text{ lbm}} = 88.44\%$$